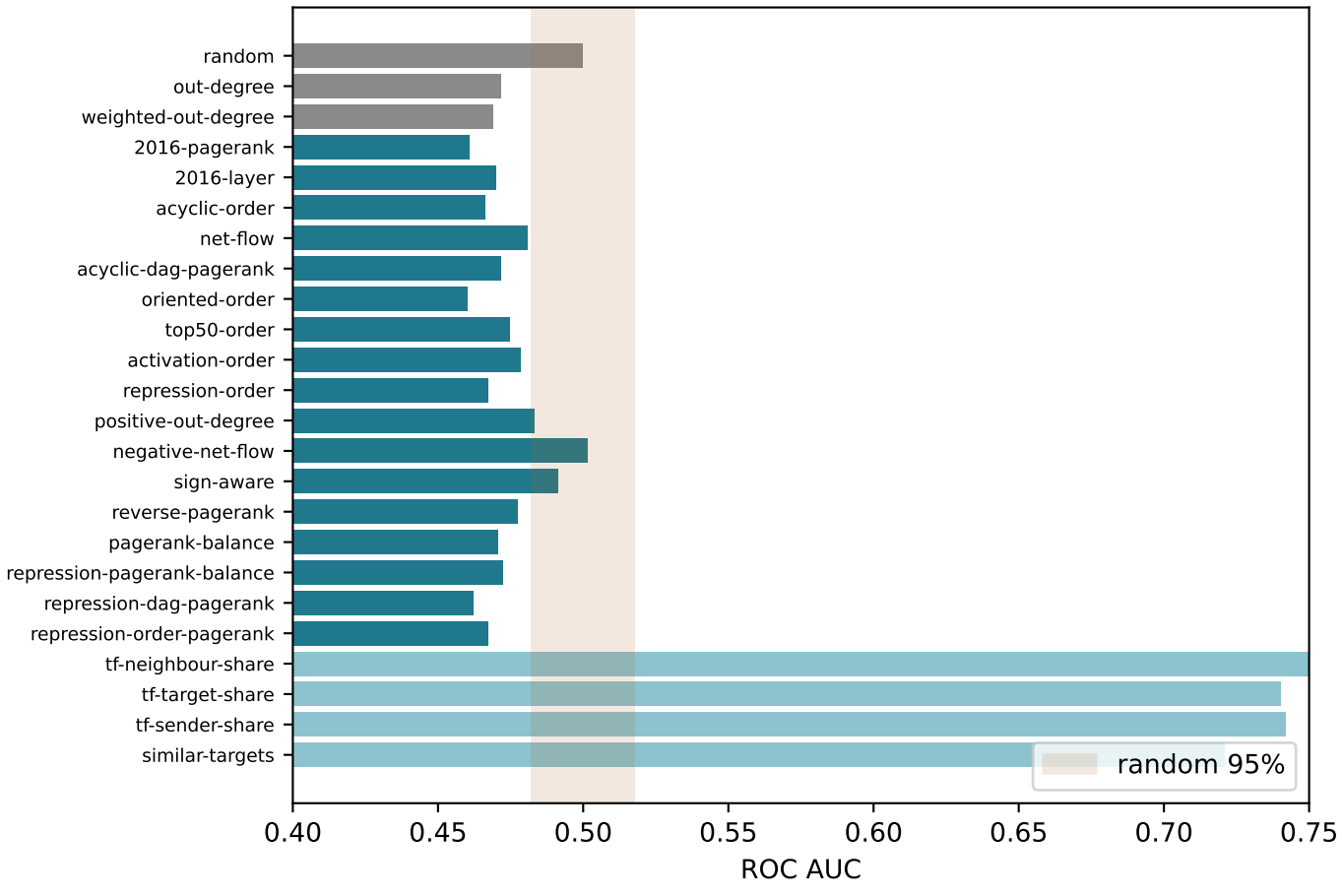
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,938 genes ranked, 1,143 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	9.5 (4–15)	0.500 (0.482–0.517)				
out-degree	6	0.472	0.927	1.000	+0.000 (+0.000, +0.000)	
weighted-out-degree	6	0.469	0.927	1.000	-0.003 (-0.004, -0.001)	< 0.001
2016-pagerank	9	0.461	0.630	1.000	-0.011 (-0.029, +0.009)	0.250
2016-layer	4	0.470	0.986	1.000	-0.002 (-0.019, +0.017)	0.886
acyclic-order	6	0.466	0.927	1.000	-0.005 (-0.012, +0.001)	0.080
net-flow	6	0.481	0.927	0.981	+0.009 (-0.002, +0.020)	0.098
acyclic-dag-pagerank	6	0.472	0.927	1.000	+0.000 (-0.005, +0.005)	0.966
oriented-order	6	0.460	0.927	1.000	-0.012 (-0.023, -0.001)	0.042
top50-order	9	0.475	0.630	0.998	+0.003 (-0.014, +0.022)	0.678
activation-order	8	0.478	0.757	0.993	+0.007 (-0.004, +0.017)	0.208
repression-order	6	0.467	0.927	1.000	-0.005 (-0.025, +0.018)	0.722
positive-out-degree †	7	0.483	0.853	0.966	+0.012 (+0.004, +0.019)	0.002
negative-net-flow †	7	0.502	0.853	0.438	+0.030 (+0.013, +0.048)	0.002
sign-aware †	6	0.491	0.927	0.824	+0.020 (+0.011, +0.029)	< 0.001
reverse-pagerank †	7	0.478	0.853	0.994	+0.006 (+0.001, +0.010)	0.014
pagerank-balance †	6	0.471	0.927	1.000	-0.001 (-0.015, +0.014)	0.906
repression-pagerank-balance †	6	0.472	0.927	1.000	+0.001 (-0.014, +0.017)	0.888
repression-dag-pagerank †	9	0.462	0.630	1.000	-0.009 (-0.029, +0.011)	0.388
repression-order-pagerank †	5	0.467	0.968	1.000	-0.005 (-0.025, +0.018)	0.732
tf-neighbour-share ‡	88	0.751	< 0.001	< 0.001	+0.279 (+0.258, +0.301)	< 0.001
tf-target-share ‡	68	0.740	< 0.001	< 0.001	+0.269 (+0.247, +0.292)	< 0.001
tf-sender-share ‡	81	0.742	< 0.001	< 0.001	+0.270 (+0.248, +0.292)	< 0.001
similar-targets ‡	81	0.721	< 0.001	< 0.001	+0.249 (+0.225, +0.274)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	31,820	0.031 (2.28×)	3.13×	0.62×	0.82 (0.79)
resolve_reciprocals → drop_sinks → prune_triangles	31,811	0.004 (0.77×)	3.65×	0.56×	0.40 (0.37)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	9,848	0.006 (0.59×)	3.75×	0.56×	0.26 (0.24)
absolute	31,820	0.031 (2.28×)			
absolute → minimum_feedback_order	31,568	0.020 (1.92×)			
orient_reciprocals ratio=2.0 → absolute	31,629	0.016 (1.77×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	31,614	0.013 (1.53×)			
top_outgoing k=50 → absolute	31,820	0.016 (4.58×)			
top_outgoing k=50 → absolute → minimum_feedback_order	31,570	0.011 (4.66×)			
keep_sign sign=1 → absolute	31,811	0.028 (3.13×)			
keep_sign sign=1 → absolute → minimum_feedback_order	31,647	0.021 (3.22×)			
keep_sign sign=-1 → absolute	20,654	0.004 (0.55×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	18,191	0.004 (0.51×)			
keep_sign sign=-1	20,654	0.004 (0.55×)		0.55×	0.26 (0.26)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 2,165,698 edges (824,160 negative), largest strong component 11,937.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 2,165,698 edges (824,160 negative), largest strong component 11,937.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 833,045 edges (790,533 negative), largest strong component 11,815.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 833,045 edges (790,533 negative), largest strong component 11,815; DAG keeps 663,658 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,165,698 edges (0 negative), largest strong component 11,937; DAG keeps 1,673,670 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 2,165,698 edges (0 negative), largest strong component 11,937.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 2,165,698 edges (0 negative), largest strong component 11,937; DAG keeps 1,673,670 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,480,108 edges (0 negative), largest strong component 11,858; DAG keeps 1,361,843 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 508,799 edges (0 negative), largest strong component 11,931; DAG keeps 350,254 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,341,538 edges (0 negative), largest strong component 11,835; DAG keeps 952,366 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 824,160 edges (0 negative), largest strong component 6,644; DAG keeps 793,586 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 2,165,698 edges (824,160 negative), largest strong component 11,937.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 2,165,698 edges (824,160 negative), largest strong component 11,937.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 2,165,698 edges (824,160 negative), largest strong component 11,937.

reverse-pagerank: Highest reverse PageRank (raw graph)